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Information processing device and information processing method

Abstract

Provided is an information processing device including at least one processor which executes an obtaining process for obtaining sensor data transmitted from each of sensors included in a sensor group, a converting process for converting the sensor data, which has been obtained in the obtaining process, into a particular format, and a providing process for providing, to main equipment, the sensor data whose format has been converted in the converting process.

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Background/Summary

(1) This Nonprovisional application claims priority under 35 U.S.C. § 119 on Patent Application No. 2021-011237 filed in Japan on Jan. 27, 2021 and on Patent Application No. 2021-171935 filed in Japan on Oct. 20, 2021, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to an information processing device and an information processing method.

BACKGROUND ART

(3) In a production site, a technique is used in which a programmable logic controller (PLC) incorporated in equipment is operated based on sensor data obtained from a plurality of sensors. As a form of connecting the plurality of sensors to the equipment, a form of connecting the plurality of sensors to the equipment via a relay device is known, as well as a form of connecting the plurality of sensors directly to the equipment.

(4) For example, Patent Literature 1 discloses a wiring system in which a plurality of proximity sensors are connected to a PLC via a relay box. The relay box disclosed in Patent Literature 1 includes a transmission wiring connection terminal connected to each of the plurality of proximity sensors, and a bus wiring connection terminal connected to the PLC. Signals outputted from the proximity sensors are collected in the relay box via the transmission wiring and are provided to the PLC via the bus wiring.

CITATION LIST

Patent Literature

(5) [Patent Literature 1]

(6) Japanese Patent Application Publication, Tokukai, No. 2011-164966

SUMMARY OF INVENTION

Technical Problem

(7) A plurality of sensors manufactured by various companies may be connected to main equipment. Moreover, sensors connected to the main equipment are not limited to a sensor (such as a proximity sensor) that transmits binary sensor data. Therefore, sensor data that the main equipment obtains from each sensor may include sensor data in a format in which the PLC of the main equipment cannot correctly interpret content of the data. Therefore, in order to utilize sensor

data in the PLC of the main equipment, the PLC of the main equipment needed to execute a converting process to convert the format of the sensor data to a format in which the PLC of the main equipment can correctly interpret content of the data. Therefore, each time a new sensor is introduced, it was necessary to prepare a program to execute the converting process corresponding to a format of sensor data transmitted from that sensor, and to install the program in the PLC of the main equipment. As a result, introduction of a new sensor was a heavy burden for users.

(8) In the wiring system disclosed in Patent Literature 1, the signal obtained from the proximity sensor is converted into a signal format of a network signal that can be transmitted by the relay box via the bus wiring. However, in this conversion, the sensor data is converted into a format that can be transmitted via the bus wiring, and not into a format in which the PLC of the main equipment can correctly interpret content of the data. Thus, sensors usable in the wiring system disclosed in Patent Literature 1 are limited to sensors that do not require the latter conversion, e.g., a proximity sensor that transmits binary sensor data.

(9) An object of an aspect of the present invention is to provide an information processing device which functions as a relay device for relaying communication between a sensor and main equipment, and which makes it possible to reduce a burden that can be caused to a user when a new sensor is introduced.

Solution to Problem

(10) In order to attain the object, an information processing device in accordance with an aspect of the present invention includes at least one processor, the at least one processor executing an obtaining process for obtaining sensor data transmitted from each of sensors included in a sensor group, a converting process for converting the sensor data, which has been obtained in the obtaining process, into a particular format that is interpretable to a controller of main equipment, and a providing process for providing, to the main equipment, the sensor data whose format has been converted in the converting process.

Advantageous Effects of Invention

(11) According to an aspect of the present invention, it is possible to provide the information processing device capable of reducing a burden on a user at the time of set-up.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a front view illustrating an appearance of an information processing device in accordance with an embodiment of the present invention.

(2) FIG. 2 is a block diagram illustrating a configuration of the information processing device illustrated in FIG. 1 and peripheral devices thereof.

(3) FIG. 3 is a diagram illustrating a configuration of sensor data stored in a memory of the information processing device illustrated in FIG. 1.

(4) FIG. 4 is a diagram illustrating a configuration of a sensor address map stored in the memory of the information processing device illustrated in FIG. 1.

(5) FIG. 5 is a diagram illustrating a configuration of equipment data stored in the memory of the information processing device illustrated in FIG. 1.

(6) FIG. 6 is a diagram illustrating a configuration of an equipment address map of equipment data stored in the memory of the information processing device illustrated in FIG. 1.

(7) FIG. 7 is a flowchart showing a method for relaying sensor data by the information processing device illustrated in FIG. 1.

(8) FIG. 8 is a flowchart showing a method for relaying equipment data by the information processing device illustrated in FIG. 1.

(9) FIG. 9 is a diagram illustrating an example of a sensor display screen which the information

processing device causes a display device to display in the method for relaying sensor data shown in FIG. 7.

(10) FIG. 10 is a diagram illustrating an example of an equipment display screen which the information processing device causes a display device to display in the method for relaying equipment data shown in FIG. 8.

DESCRIPTION OF EMBODIMENTS

(11) An information processing device in accordance with the present application is a device for relaying communication between sensors included in a sensor group and main equipment, and relaying communication between pieces of accessory equipment included in an accessory equipment group and the main equipment. The following description will discuss details of an information processing device in accordance with the present embodiment with reference to the drawings.

(12) (Appearance of Information Processing Device)

(13) The following description will discuss an appearance of an information processing device 1 in accordance with an embodiment of the present invention with reference to FIG. 1. FIG. 1 is a front view illustrating an appearance of the information processing device 1.

(14) As illustrated in FIG. 1, the information processing device 1 includes a power source button 7, a power source pilot lamp 8, and an abnormality pilot lamp 9.

(15) The power source button 7 is a button for starting or stopping the information processing device 1. When a user presses the power source button 7 in a state in which the information processing device 1 is stopped, the information processing device 1 starts operating. When the user presses the power source button 7 in a state in which the information processing device 1 is in operation, the information processing device 1 stops. The power source pilot lamp 8 is a pilot lamp for indicating that the information processing device 1 is in operation. In a state in which the information processing device 1 is in operation, the power source pilot lamp 8 is on. In a state in which the information processing device 1 is stopped, the power source pilot lamp 8 is off. The abnormality pilot lamp 9 is a pilot lamp for indicating that an abnormality is occurring in the information processing device 1. In a state in which the information processing device 1 is in normal operation, the abnormality pilot lamp 9 is off. In a state in which the information processing device 1 is not in normal operation, the abnormality pilot lamp 9 is on. The abnormality pilot lamp 9 can be configured to be turned on when an abnormality occurs in the sensor or the accessory equipment that is connected to the information processing device 1.

(16) (Configuration of Information Processing System Including Information Processing Device)

(17) The following description will discuss a configuration of an information processing system S including the information processing device 1 with reference to FIG. 2. FIG. 2 is a block diagram illustrating the configuration of the information processing system S.

(18) The information processing system S includes main equipment 2, a sensor group 3, an accessory equipment group 4, and a display device 5 in addition to the information processing device 1.

(19) The main equipment 2 is a device which plays a main role in a production line. The main equipment 2 can be, for example, a molding machine for forming a mold. The main equipment 2 includes a programmable logic controller (PLC) 21. The PLC 21 controls components of the main equipment 2 based on sensor data obtained, via the information processing device 1, from sensors constituting the sensor group 3 and based on pieces of equipment data obtained, via the information processing device 1, from pieces of accessory equipment constituting the accessory equipment group 4.

(20) The sensor group 3 includes a plurality of sensor slave units 3a1, 3a2, . . . (an example of “sensors” in claims) and a sensor master unit 3b. Hereinafter, when the sensor slave units 3a1, 3a2, . . . are not individually identified, the sensor slave unit is referred to as “sensor slave unit 3a”.

(21) The sensor slave unit 3a is a device for detecting surrounding statuses of the sensor slave unit

3a, e.g., a status of the main equipment **2**, a surrounding status of the main equipment **2**, a status of accessory equipment **4a**, a surrounding status of the accessory equipment **4a**, a status of a raw material, a status of an intermediate product, or a status of a product. The sensor slave unit **3a** can be, for example, an open/close sensor, a temperature sensor, PATLITE (registered trademark), or the like. The sensor slave unit **3a** transmits, to the sensor master unit **3b**, a detection value of the sensor slave unit **3a** along with status information of the sensor slave unit **3a** as sensor data. Here, the status information of the sensor slave unit **3a** indicates, for example, a usage status of the sensor slave unit **3a**, presence or absence of a monitoring function, a communication status, radio field intensity, or a remaining battery capacity. The sensor slave unit **3a** is wirelessly connected to the sensor master unit **3b** via, for example, EnOcean (registered trademark), ZigBee (registered trademark), Bluetooth Low Energy (registered trademark), or the like.

(22) Note that there is a sensor device which has a plurality of sensor functions, e.g., a temperature/humidity sensor which detects both a temperature and humidity. In this case, each of sensor functions of a sensor device is dealt with as the sensor slave unit **3a**. Therefore, a plurality of sensor slave units **3a** can correspond to a single sensor device.

(23) The sensor master unit **3b** is a device for relaying communication between each of the sensor slave units **3a** and the information processing device **1**. The sensor master unit **3b** transmits, to the information processing device **1**, detection values and status information of the sensor slave units **3a** that are connected to the sensor master unit **3b** along with status information of the sensor master unit **3b** as sensor data. Here, the status information of the sensor master unit **3b** indicates, for example, a communication status and the number of times of data obtainment of the sensor master unit **3b**. The sensor master unit **3b** is connected to the information processing device **1** via wired connection by, for example, Ethernet (registered trademark).

(24) The accessory equipment group **4** is constituted by a plurality of pieces of accessory equipment **4a1**, **4a2**, and so forth. Hereinafter, when the pieces of accessory equipment **4a1**, **4a2**, . . . are not individually identified, the accessory equipment is referred to as “accessory equipment **4a**”.

(25) The accessory equipment **4a** is a device for assisting the main equipment **2**. The accessory equipment **4a** can be, for example, a conveyor, a cylinder, a dust collector, or the like. The accessory equipment **4a** includes a PLC **41**. The PLC **41** of each of pieces of accessory equipment **4a** controls components of that accessory equipment **4a**. The PLC **41** of each of the pieces of accessory equipment **4a** transmits, to the information processing device **1**, digital output and analog output of that accessory equipment **4a** along with status information of that accessory equipment **4a** as equipment data. Here, the digital output of the accessory equipment **4a** is also called “operation signal” and indicates, for example, whether or not automatic operation is available, whether or not the equipment is in automatic operation, whether or not an abnormality is occurring, or whether or not the equipment is in emergency stop. The analog output of the accessory equipment **4a** indicates, for example, a feeding speed of a conveyor, operation pressure of a cylinder, an airflow rate of a dust collector, power consumption, or the like. The status information of the accessory equipment **4a** indicates, for example, a usage status of the accessory equipment **4a**, presence or absence of a monitoring function, or a communication status. The accessory equipment **4a** is connected to the information processing device **1** via wired connection by, for example, CC-Link (registered trademark).

(26) The information processing device **1** is a device for relaying communication between the main equipment **2** and the sensor group **3**, and communication between the main equipment **2** and the accessory equipment group **4**. The information processing device **1** includes a processor **11**, a memory **12**, an input/output IF **13**, a communication IF **14**, and a PLC **15**, as illustrated in FIG. 2. The processor **11**, the memory **12**, the input/output IF **13**, the communication IF **14**, and the PLC **15** are connected to each other via buses. The information processing device **1** can be realized, for example, by adding the PLC **15** to a small single board computer such as Raspberry Pi (registered

trademark).

(27) In the memory **12**, a sensor data set **D1**, an equipment data set **D2**, a sensor address map **M1**, and an equipment address map **M2** are stored together with an information processing program **P**. The processor **11** refers to the sensor data set **D1**, the equipment data set **D2**, the sensor address map **M1**, and the equipment address map **M2**, and executes an information processing method (described later) in accordance with the information processing program **P**. Examples of devices that can be used as the processor **11** include a central processing unit (CPU). Examples of devices that can be used as the memory **12** include a flash memory.

(28) The input/output **IF 13** is an interface for connecting the display device **5**. As the input/output **IF 13**, for example, high-definition multimedia interface (HDMI) (registered trademark) is used. The communication **IF 14** is an interface for communicating with the sensor master unit **3b**. For example, an Ethernet interface is used as the communication **IF 14**. The PLC **15** is a controller for communicating with (the PLC **21** of) the main equipment **2** and (the PLC **41** of) the accessory equipment **4a**. The PLC **15** can access the equipment data set **D2** and the equipment address map **M2** via the busses. The PLC **15** is connected to (the PLC **21** of) the main equipment **2** and to (the PLC **41** of) the accessory equipment **4a** by CC-Link.

(29) (Configurations of Sensor Data Set and Sensor Address Map)

(30) A configuration of the sensor data set **D1** stored in the memory **12** of the information processing device **1** will be described with reference to FIG. **3**. FIG. **3** is a diagram illustrating the configuration of the sensor data set **D1**.

(31) As illustrated in FIG. **3**, the sensor data set **D1** is constituted by configuration information **91**, status information **92**, output information **93**, and monitor information **94**. Pieces of information constituting the configuration information **91**, the status information **92**, the output information **93**, and the monitor information **94** are managed with consecutive word numbers. In FIG. **3**, a leading portion of word number of the configuration information **91** is “Head”, a leading portion of word number of the status information **92** is “Comm.”, a leading portion of word number of the output information **93** is “Data”, and a leading portion of word number of the monitor information **94** is “Mon.”, and each of the leading portions indicates content of information corresponding to that word number.

(32) The configuration information **91** is information indicating a configuration of the sensor group **3**. The configuration information **91** includes, for example, the number of sensor master units **3b**, a type code and an IP address of each sensor master unit **3b**, the number of sensor slave units **3a** connected to each sensor master unit **3b**, the total number of sensor slave units **3a**, and a type code of each sensor slave unit **3a**. For example, the configuration information **91** may be set by a manufacturer prior to shipment of the information processing device **1**, or may be set by a user after shipment of the information processing device **1**. An offset value of a leading word number of each of the status information **92**, the output information **93**, and the monitor information **94** (i.e., a difference from a leading word number of the configuration information **91**), system information of the information processing device **1**, and the like may be included in the configuration information **91**.

(33) The status information **92** is a set of status information of each of the sensor slave units **3a** and status information of the sensor master unit **3b**. The status information of each of the sensor slave units **3a** is, as described above, obtained from that sensor slave unit **3a** via the sensor master unit **3b** as sensor data. The status information of the sensor master unit **3b** is obtained from the sensor master unit **3b** as described above. The status information **92** may include a keep-alive signal of the information processing system **S**, sensor slave unit integrated information in which pieces of status information of all the sensor slave units **3a** are integrated, and/or the like. The sensor slave unit integrated information includes, for example, information indicating whether or not communication statuses of all the sensor slave units **3a** are normal, and information indicating whether or not remaining battery capacities of all the sensor slave units **3a** are normal.

- (34) The output information **93** is a set of detection values of the sensor slave units **3a**. The detection value of each of the sensor slave units **3a** is obtained as sensor data from that sensor slave unit **3a**, as described above.
- (35) The monitor information **94** is a set of a determination condition and a determination result pertaining to each of the sensor slave units **3a**. The determination condition pertaining to each of the sensor slave units **3a** is represented, for example, by a threshold value. In this case, the determination result pertaining to each of the sensor slave units **3a** is given as a result of comparing the detection value of that sensor slave unit **3a** with the threshold value. The determination condition pertaining to each of the sensor slave units **3a** may be set by a manufacturer prior to shipment of the information processing device **1**, or may be set by a user after shipment of the information processing device **1**. The determination result pertaining to each of the sensor slave units **3a** is derived from the information processing device **1** each time the detection value of that sensor slave unit **3a** is updated.
- (36) A configuration of the sensor address map **M1** as address information stored in the memory **12** of the information processing device **1** will be described with reference to FIG. 4. FIG. 4 is a diagram illustrating the configuration of the sensor address map **M1**.
- (37) The sensor address map **M1** is a table in which word numbers assigned to respective pieces of information included in the sensor data set **D1** are associated with addresses on a memory of the PLC **21** provided in the main equipment **2**.
- (38) The processor **11** of the information processing device **1** reads out pieces of information included in the sensor data set **D1** as the configuration information **91**, the status information **92**, the output information **93**, and the monitor information **94** from the memory **12** of the information processing device **1**, and writes the information to the memory of the PLC **21** provided in the main equipment **2**. At this time, the processor **11** of the information processing device **1** determines, in accordance with the sensor address map **M1**, a region on the memory of the PLC **21** to which each piece of information is to be written. For example, in the sensor address map **M1**, a word number Data+001 is associated with addresses EEE through E'E'E'. Therefore, the processor **11** of the information processing device **1** writes detection values of a first sensor slave unit, to which the word number Data+001 is assigned in the sensor data set **D1**, to regions to which the addresses EEE through E'E'E' are assigned in the memory of the PLC **21**.
- (39) The processor **11** of the information processing device **1** may write sensor data obtained from the sensors constituting the sensor group **3** to the memory of the PLC **21** provided in the main equipment **2** in accordance with the information processing program **P**. In this case, address information equivalent to the foregoing sensor address map **M1** is included inside the information processing program **P**.
- (40) (Configurations of Equipment Data Set and Equipment Address Map)
- (41) A configuration of the equipment data set **D2** stored in the memory **12** of the information processing device **1** will be described with reference to FIG. 5. FIG. 5 is a diagram illustrating the configuration of the equipment data set **D2**.
- (42) As illustrated in FIG. 5, the equipment data set **D2** is constituted by configuration information **95**, status information **96**, output information **97**, input information **98**, and monitor information **99**. Pieces of information constituting the configuration information **95**, the status information **96**, the output information **97**, the input information **98**, and the monitor information **99** are managed with consecutive word numbers. In FIG. 5, a leading portion of word number of the configuration information **95** is "Head", a leading portion of word number of the status information **96** is "Comm.", a leading portion of word number of the output information **97** is "S_data", a leading portion of word number of the input information **98** is "O_data", and a leading portion of word number of the monitor information **99** is "Mon.", and each of the leading portions indicates content of information corresponding to that word number.
- (43) The configuration information **95** is information indicating a configuration of the accessory

equipment group **4**. The configuration information **95** includes, for example, the number of pieces of the accessory equipment **4a** and a type code of each piece of the accessory equipment **4a**. An offset value of a leading word number of each of the status information **96**, the output information **97**, the input information **98**, and the monitor information **99** (i.e., a difference from a leading word number of the configuration information **95**), system information of the information processing device **1**, and the like may be included in the configuration information **95**.

(44) The status information **96** is a set of pieces of status information of the pieces of accessory equipment **4a**. The status information of each piece of the accessory equipment **4a** is obtained as the equipment data from that piece of accessory equipment **4a**, as described above. The status information **96** may include a keep-alive signal of the information processing system **S**, a low battery signal of the information processing device **1**, and/or the like.

(45) The output information **97** is a set of digital output and analogue output of each piece of the accessory equipment **4a**. The digital output and analogue output of each piece of the accessory equipment **4a** are obtained as the equipment data from that piece of accessory equipment **4a**, as described above. The output information **97** may include accessory equipment integrated information in which digital outputs (operation signals) of all pieces of the accessory equipment **4a** are integrated. The accessory equipment integrated information includes, for example, information indicating whether or not all pieces of accessory equipment **4a** are normal, and information indicating whether or not all pieces of the accessory equipment **4a** can be automatically operated.

(46) The input information **98** is a set of digital input and analog input of each piece of the accessory equipment **4a**. The digital input of each piece of the accessory equipment **4a** is also called “command signal”, and represents a command to that piece of accessory equipment **4a**. The analog input of each piece of the accessory equipment **4a** is also called “command value”, and represents a command value to that piece of accessory equipment **4a**. The digital input and analog input of each piece of the accessory equipment **4a** are obtained from the PLC **21** of the main equipment **2**. The input information **98** may include an accessory equipment integrated command in which command signals for all pieces of the accessory equipment **4a** are integrated. The accessory equipment integrated command includes, for example, an auto-start command for all pieces of accessory equipment that gives an instruction on auto-start to all pieces of the accessory equipment **4a**, and an auto-stop command for all pieces of accessory equipment that gives an instruction on auto-stop to all pieces of the accessory equipment **4a**.

(47) The monitor information **99** is a set of a determination condition and a determination result pertaining to each piece of the accessory equipment **4a**. The determination condition pertaining to each piece of the accessory equipment **4a** is represented, for example, by a threshold value. In this case, the determination result pertaining to each piece of the accessory equipment **4a** is given as a result of comparing the digital output of that piece of accessory equipment **4a** with the threshold value. The determination condition pertaining to each piece of the accessory equipment **4a** may be set by a manufacturer prior to shipment of the information processing device **1**, or may be set by a user after shipment of the information processing device **1**. The determination result pertaining to each piece of the accessory equipment **4a** is derived by the information processing device **1** each time the digital output of that piece of accessory equipment **4a** is updated. A determination record pertaining to each piece of the accessory equipment **4a** may be included in the monitor information **99**. The determination record pertaining to each piece of the accessory equipment **4a** includes, for example, the number of times the determination result pertaining to that piece of accessory equipment **4a** has become true.

(48) A configuration of the equipment address map **M2** as address information stored in the memory **12** of the information processing device **1** will be described with reference to FIG. **6**. FIG. **6** is a diagram illustrating the configuration of the equipment address map **M2**.

(49) The equipment address map **M2** is a table in which word numbers assigned to respective pieces of information included in the equipment data set **D2** are associated with addresses on a

memory of the PLC **21** provided in the main equipment **2**.

(50) The processor **11** of the information processing device **1** reads out pieces of information included in the equipment data set **D2** as the configuration information **95**, the status information **96**, the output information **97**, and the monitor information **99** from the memory **12** of the information processing device **1**, and writes the information to the memory of the PLC **21** provided in the main equipment **2**. At this time, the processor **11** of the information processing device **1** determines, in accordance with the equipment address map **M2**, a region on the memory of the PLC **21** to which each piece of information is to be written. For example, in the equipment address map **M2**, a word number **S_S_Data+001** is associated with addresses **UUU** through **U'U'U'**. Therefore, the processor **11** of the information processing device **1** writes digital output of first accessory equipment, to which the word number **S_Data+001** is assigned in the equipment data set **D2**, to regions to which the addresses **UUU** through **U'U'U'** are assigned in the memory of the PLC **21**.

(51) In addition, the processor **11** of the information processing device **1** reads out pieces of information included in the equipment data set **D2** as the input information **98** from the memory of the PLC **21** provided in the main equipment **2**, and writes the read information to the memory **12** of the information processing device **1**. At this time, the processor **11** of the information processing device **1** determines, in accordance with the equipment address map **M2**, a region on the memory of the PLC **21** from which each piece of information is to be read out. For example, in the equipment address map **M2**, a word number **O_Data+001** is associated with addresses **WWW** through **W'W'W'**. Therefore, the processor **11** of the information processing device **1** reads out digital input of first accessory equipment, to which the word number **O_Data+001** is assigned in the sensor data set **D1**, from regions to which the addresses **WWW** through **W'W'W'** are assigned in the memory of the PLC **21**.

(52) The processor **11** of the information processing device **1** may write pieces of equipment data obtained from the pieces of accessory equipment constituting the accessory equipment group **4** to the memory of the PLC **21** provided in the main equipment **2** in accordance with the information processing program **P**. In this case, address information corresponding to the foregoing equipment address map **M2** is included inside the information processing program **P**.

(53) (Sensor Data Relaying Method)

(54) A method **Sa** for relaying sensor data in the information processing device **1** will be described with reference to FIG. 7. FIG. 7 is a flowchart showing a method **Sa** for relaying sensor data.

(55) The method **Sa** for relaying sensor data includes an obtaining process **S1**, a converting process **S2**, a determining process **S3**, a providing process **S4**, and a displaying process **S5**, as shown in FIG. 7.

(56) The obtaining process **S1** is a process in which the processor **11** of the information processing device **1** obtains sensor data transmitted from the sensor slave units **3a** constituting the sensor group **3**. In the obtaining process **S1**, the sensor slave unit **3a** transmits sensor data to the sensor master unit **3b** periodically or each time a detection value changes. Further, the sensor master unit **3b** transmits the sensor data received from the sensor slave unit **3a** to the information processing device **1** periodically or each time the sensor master unit **3b** receives sensor data from the sensor slave unit **3a**. The processor **11** of the information processing device **1** receives the sensor data transmitted from the sensor slave unit **3a**.

(57) The converting process **S2** is a process for converting sensor data obtained in the obtaining process **S1** into a particular format that is interpretable to the PLC **21** of the main equipment **2**. In the converting process **S2**, the processor **11** of the information processing device **1** extracts status information and a detection value of the sensor slave unit **3a** from sensor data obtained in the obtaining process **S1**. In addition, the processor **11** of the information processing device **1** converts the extracted status information into a format that is interpretable to the PLC **21** of the main equipment **2**, and then writes the converted information to the memory **12** as the status information **92** of the sensor data set **D1**. The processor **11** of the information processing device **1** converts the

extracted detection value into a format that is interpretable to the PLC **21** of the main equipment **2**, and then writes the converted value to the memory **12** as the output information **93** of the sensor data set **D1**. For example, in a case where sensor data of “55:0:07:02:0a:0a:21:05:1c:fc:a6:09:b8:01:37:90” is obtained from a certain sensor, the processor **11** converts information pertaining to the detection value in the obtained sensor data into “1” and converts information pertaining to the status information in the obtained sensor data into “RSSI-55”.

(58) The determining process **S3** is a process for determining whether or not the sensor data obtained in the obtaining process **S1** satisfies a particular condition. In the determining process **S3**, the processor **11** of the information processing device **1** (1) compares the detection value of the sensor slave unit **3a** written to the memory **12** as the output information **93** of the sensor data set **D1** with the threshold value written to the memory **12** as the monitor information **94** of the sensor data set **D1**, and (2) writes a comparison result to the memory **12** as the monitor information **94** of the sensor data set **D1**.

(59) The providing process **S4** is a process for providing, to the main equipment **2**, sensor data whose format has been converted in the converting process **S2**. In the providing process **S4**, the processor **11** of the information processing device **1** reads out pieces of information constituting the status information **92**, the output information **93**, and the monitor information **94** of the sensor data set **D1**, and writes the read information to the memory of the PLC **21** provided in the main equipment **2**. At this time, the processor **11** of the information processing device **1** determines, with reference to the sensor address map **M1**, which information is to be written to which region of the memory of the PLC **21**.

(60) The displaying process **S5** is a process for displaying, on the display device **5**, sensor data obtained in the obtaining process **S1** or information obtained from the sensor data. In the displaying process **S5**, the processor **11** of the information processing device **1** generates a sensor display screen for displaying sensor data. The processor **11** of the information processing device **1** outputs the generated sensor display screen to the display device **5**. A specific example of the sensor display screen will be described later with reference to another drawing.

(61) As illustrated in FIG. 7, the processor **11** of the information processing device **1** may further execute a changing process **S6** for changing a condition used for the determining process **S3** based on a user operation. In the changing process **S6**, the processor **11** of the information processing device **1** changes the threshold value stored in the memory **12** as the monitor information **94** of the sensor data set **D1** to a value specified by the user.

(62) The processor **11** of the information processing device **1** may further execute a monitoring process **S7**, a notifying process **S8**, and a reconnecting process **S9**, as illustrated in FIG. 7.

(63) The monitoring process **S7** is a process in which the processor **11** of the information processing device **1** monitors a data transmission interval of each of the sensor slave units **3a** included in the sensor group **3**. In the monitoring process **S7**, the processor **11** of the information processing device **1** monitors intervals of data transmission from the sensor master unit **3b**. The processor **11** monitors the data transmission interval based on the status information **92** of the sensor data set **D1** stored in the memory **12**. In a case where the data transmission interval of the sensor master unit **3b** exceeds a particular period, the processor **11** determines that connection with the sensor master unit **3b** is erroneous.

(64) The notifying process **S8** is a process for giving notification of a connection error in a case where the data transmission interval of the sensor master unit **3b** monitored in the monitoring process **S7** exceeds the particular period. In a case where it is determined in the monitoring process **S7** that connection with the sensor master unit **3b** is erroneous, the processor **11** of the information processing device **1** causes the display device **5** to display notification of a connection error in the notifying process **S8**.

(65) The reconnecting process **S9** is a process in which the processor **11** of the information

processing device **1** attempts to reconnect to the sensor master unit **3b** when communication with the sensor master unit **3b** has been shut off. In the reconnecting process **S9**, the processor **11** of the information processing device **1** executes reconnection to the sensor master unit **3b** for which a connection error has been determined in the monitoring process **S7**. The processor **11** of the information processing device **1** repeats the reconnecting process **S9** until the processor **11** is connected to the sensor master unit **3b**.

(66) (Equipment Data Relaying Method)

(67) A method **Sb** for relaying equipment data in the information processing device **1** will be described with reference to FIG. **8**. FIG. **8** is a flowchart showing a method **Sb** for relaying equipment data.

(68) The method **Sb** for relaying equipment data includes an obtaining process **S20**, a converting process **S21**, a determining process **S22**, a providing process **S23**, and a displaying process **S24**, as shown in FIG. **8**.

(69) The obtaining process **S20** is a process in which the processor **11** of the information processing device **1** obtains pieces of equipment data transmitted from the pieces of accessory equipment **4a** constituting the accessory equipment group **4**. In the obtaining process **S20**, a piece of accessory equipment **4a1** transmits equipment data to the information processing device **1** periodically or each time the status changes.

(70) The converting process **S21** is a process for converting equipment data obtained in the obtaining process **S20** into a particular format that is interpretable to the PLC **21** of the main equipment **2**. In the converting process **S21**, the processor **11** of the information processing device **1** extracts status information, digital output and analogue output of the piece of accessory equipment **4a1** from the equipment data obtained in the obtaining process **S20**. In addition, the processor **11** of the information processing device **1** converts the extracted status information into a format that is interpretable to the PLC **21** of the main equipment **2**, and then writes the converted information to the memory **12** as the status information **96** of the equipment data set **D2**. In addition, the processor **11** of the information processing device **1** converts the extracted digital output and analog output into a format that is interpretable to the PLC **21** of the main equipment **2**, and then writes the converted output to the memory **12** as the output information **97** of the equipment data set **D2**. For example, the processor **11** converts equipment data into a particular format by executing a process of sorting information included in the equipment data (sorting) and/or a process of gathering information included in the equipment data (bit packing). In the method **Sb** for relaying equipment data, it is possible that the converting process **S21** is omitted, and the equipment data is written in the memory **12** without converting the format.

(71) The determining process **S22** is a process for determining whether or not the equipment data obtained in the obtaining process **S20** satisfies a particular condition. In the determining process **S22**, the processor **11** of the information processing device **1** (1) compares the digital output and/or analog output of the piece of accessory equipment **4a1** written to the memory **12** as the output information **97** of the equipment data set **D2** with the threshold value written to the memory **12** as the monitor information **99** of the equipment data set **D2**, and (2) writes a comparison result to the memory **12** as the monitor information **99** of the equipment data set **D2**.

(72) The providing process **S23** is a process for providing, to the main equipment **2**, equipment data whose format has been converted in the converting process **S21**. In the providing process **S23**, the processor **11** of the information processing device **1** reads out information constituting the status information **96**, the output information **97**, and the monitor information **99** of the equipment data set **D2**, and writes the read information to the memory of the PLC **21** provided in the main equipment **2**. At this time, the processor **11** of the information processing device **1** determines, with reference to the equipment address map **M2**, which information is to be written to which region of the memory of the PLC **21**.

(73) The displaying process **S24** is a process for displaying, on the display device **5**, equipment

data obtained in the obtaining process S20 or information obtained from the equipment data. In the displaying process S24, the processor 11 of the information processing device 1 generates an accessory equipment display screen for displaying equipment data. The processor 11 of the information processing device 1 outputs the generated accessory equipment display screen to the display device 5. A specific example of the accessory equipment display screen will be described later with reference to another drawing.

(74) As illustrated in FIG. 8, the processor 11 of the information processing device 1 may further execute a changing process S25 for changing a condition used for the determining process S22 based on a user operation. In the changing process S25, the processor 11 of the information processing device 1 changes the threshold value stored in the memory 12 as the monitor information 99 of the equipment data set D2 to a value specified by the user.

(75) The processor 11 of the information processing device 1 may further execute an obtaining process S26 and a providing process S27, as illustrated in FIG. 8.

(76) The obtaining process S26 is a process in which the processor 11 of the information processing device 1 obtains a command signal and a command value from the main equipment 2. In the obtaining process S26, the processor 11 of the information processing device 1 reads out the command signal and the command value from the memory of the PLC 21 of the main equipment 2. The processor 11 of the information processing device 1 writes the read command signal and command value to the memory 12 as the input information 98 of the equipment data set D2. At this time, the processor 11 of the information processing device 1 determines, with reference to the equipment address map M2, which information is to be read out from which region of the memory of the PLC 21.

(77) The providing process S27 is a process for providing the command signal and the command value to each piece of the accessory equipment 4a. In the providing process S27, the processor 11 of the information processing device 1 reads out, from the memory 12, information constituting the output information 97 of the equipment data set D2, and transmits the read information to pieces of the accessory equipment 4a.

(78) (Examples of Display Screen)

(79) Examples of display screens which the information processing device 1 causes the display device 5 to display will be described with reference to FIG. 9 and FIG. 10. FIG. 9 illustrates an example of a sensor display screen which the information processing device 1 causes the display device 5 to display in the method Sa for relaying sensor data. FIG. 10 illustrates an example of an equipment display screen which the information processing device 1 causes the display device 5 to display in the method Sb for relaying equipment data.

(80) The sensor display screen illustrated in FIG. 9 indicates information pertaining to a sensor slave unit 3a selected by the user and information pertaining to a sensor master unit 3b that relays sensor data transmitted from that sensor slave unit 3a.

(81) The information pertaining to the sensor slave unit 3a includes configuration information of the sensor slave unit 3a (name and ID in FIG. 9) and status information of the sensor slave unit 3a (remaining battery capacity and radio field intensity in FIG. 9), as well as a detection value of the sensor slave unit 3a (indicators showing temperature and vibration in FIG. 9). The information pertaining to sensor master unit 3b includes configuration information of the sensor master unit 3b (name in FIG. 9), as well as status information of the sensor master unit 3b.

(82) The equipment display screen illustrated in FIG. 10 indicates information pertaining to a piece of accessory equipment 4a selected by the user.

(83) The information pertaining to the piece of accessory equipment 4a includes monitor information of the piece of accessory equipment 4a (replacement information, setting number of times, and current values in FIG. 10), as well as an operation signal of the piece of accessory equipment 4a (indicators showing whether or not the equipment is in operation, and whether or not the equipment is in abnormal operation in FIG. 10). The equipment display screen may also include

a user interface for changing the threshold value included in the monitor information, as illustrated in FIG. 10.

Additional Remarks

(84) The present invention is not limited to the embodiments, but can be altered by a skilled person in the art within the scope of the claims. The present invention also encompasses, in its technical scope, any embodiment derived by combining technical means disclosed in differing embodiments.

REFERENCE SIGNS LIST

(85) **1:** Information processing device **2:** Main equipment **3:** Sensor group **3a:** Sensor slave unit **3b:** Sensor master unit **4:** Accessory equipment group **5:** Display device

Claims

1. An information processing device for relaying communication between sensors included in a sensor group and main equipment connected to the information processing device via a network, the information processing device comprising at least one processor, the at least one processor configured to obtain sensor data transmitted from each of the sensors included in the sensor group, convert the sensor data, which has been obtained, into a particular format that is interpretable to a controller of the main equipment, and provide, to the main equipment, the sensor data whose format has been converted, wherein the sensor group is constituted by a plurality of sensor slave units that are the sensors and a sensor master unit configured to relay communication between each of the plurality of sensor slave units and the information processing device, the sensor data includes (i) a detection value and status information of one of the plurality of sensor slave units and (ii) status information of the sensor master unit, the status information of one of the plurality of sensor slave units indicates at least one of a usage status, presence or absence of a monitoring function, a communication status, radio field intensity, and a remaining battery capacity, and the status information of the sensor master unit indicates at least one of a communication status and the number of times of data obtainment of the sensor master unit.
2. The information processing device as set forth in claim 1, further comprising a first memory storing address information in which each of the sensors included in the sensor group is associated with a region, to which the sensor data transmitted from that sensor is to be written, on a second memory of the main equipment, the at least one processor writing the sensor data, which has been transmitted from each of the sensors included in the sensor group, to the region on the second memory, the region being associated with that sensor in the address information.
3. The information processing device as set forth in claim 2, wherein the first memory stores a sensor data set, and the sensor data set is constituted by (i) configuration information indicating a configuration of the sensor group, (ii) status information that is a set of status information of each of the plurality of sensor slave units and status information of the sensor master unit, (iii) output information that is a set of detection values of the plurality of sensor slave units, and (iv) monitor information that is a set of a determination condition and a determination result pertaining to each of the plurality of sensor slave units.
4. The information processing device as set forth in claim 1, wherein the at least one processor is further configured to display, on a display device, the sensor data obtained or information obtained from the sensor data.
5. The information processing device as set forth in claim 4, wherein: the at least one processor is further configured to determine whether or not the sensor data obtained satisfies a particular condition; and the at least one processor causes the display device to display a result of whether or not the sensor data satisfies the particular condition.
6. The information processing device as set forth in claim 5, wherein the at least one processor is further configured to change the particular condition based on a user operation.
7. The information processing device as set forth in claim 1, wherein: the at least one processor is

further configured to monitor data transmission intervals of the sensors included in the sensor group; and the at least one processor is further configured to give notification of a connection error when a data transmission interval of any of the sensors included in the sensor group has exceeded a particular time interval.

8. The information processing device as set forth in claim 1, wherein: the at least one processor obtains, via the sensor master unit, the sensor data transmitted from each of the sensors included in the sensor group; and the at least one processor attempts to reconnect to the sensor master unit when communication with the sensor master unit has been shut off.

9. The information processing device according to claim 1, further comprising a programmable logic controller (PLC), the PLC being configured to communicate with the main equipment via the network.

10. The information processing device according to claim 1, wherein: the information processing device further comprises a first memory; and the at least one processor is configured to: write the sensor data, which has been converted into the particular format, to the first memory; and provide the sensor data, which has been written to the first memory, to the main equipment.

11. The information processing device according to claim 10, wherein the at least one processor writes the sensor data, which has been written to the first memory, to a second memory included in the main equipment.

12. The information processing device as set forth in claim 10, wherein the first memory stores a sensor data set, the sensor data set is constituted by (i) configuration information indicating a configuration of the sensor group, (ii) status information that is a set of status information of each of the plurality of sensor slave units and status information of the sensor master unit, (iii) output information that is a set of detection values of the plurality of sensor slave units, and (iv) monitor information that is a set of a determination condition and a determination result pertaining to each of the plurality of sensor slave units, and the processor reads out information included in the sensor data set and writes the information to a second memory of the main equipment.

13. The information processing device according to claim 1, wherein one of the plurality of sensor slave units transmits the detection value and the status information of the one of the plurality of sensor slave units as sensor data each time the detection value changes, and the sensor master unit transmits, to the information processing device, the sensor data received from the one of the plurality of sensor slave units.

14. The information processing device according to claim 1, wherein the sensor master unit is connected to the information processing device via wired connection.

15. An information processing method using an information processing device for relaying communication between a sensor group and main equipment connected to the information processing device via a network, wherein the sensor group is constituted by a plurality of sensor slave units and a sensor master unit configured to relay communication between each of the plurality of sensor slave units and the information processing device, the information processing method comprising: obtaining, by at least one processor, sensor data transmitted from the sensor master unit; converting, by the at least one processor, the sensor data, which has been obtained, into a particular format; and providing, by the at least one processor, the sensor data whose format has been converted in to the main equipment, wherein the sensor data includes (i) a detection value and status information of one of the plurality of sensor slave units which detection value and status information have been transmitted from the one of the sensor slave units to the sensor master unit and (ii) status information of the sensor master unit, the status information of one of the plurality of sensor slave units indicates at least one of a usage status, presence or absence of a monitoring function, a communication status, radio field intensity, and a remaining battery capacity, and the status information of the sensor master unit indicates at least one of a communication status and the number of times of data obtainment of the sensor master unit.

16. An information processing device for relaying communication between sensors included in a

sensor group and main equipment connected to the information processing device via a network, the information processing device comprising at least one processor, the at least one processor configured to obtain sensor data transmitted from each of sensors included in the sensor group, and convert the sensor data, which has been obtained, into a particular format that is interpretable to a controller of the main equipment, wherein the sensor group is constituted by a plurality of sensor slave units that are the sensors and a sensor master unit configured to relay communication between each of the plurality of sensor slave units and the information processing device, the sensor data includes (i) a detection value and status information of one of the plurality of sensor slave units and (ii) status information of the sensor master unit, the status information of one of the plurality of sensor slave units indicates at least one of a usage status, presence or absence of a monitoring function, a communication status, radio field intensity, and a remaining battery capacity, and the status information of the sensor master unit indicates at least one of a communication status and the number of times of data obtainment of the sensor master unit.
